THE CURVES STACK IS FIBRED IN GROUPOIDS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Equation equation-curves-over-proper-spaces (canonical tag 0D50). cited at proof line 4793. The embedding of the curves stack.
- Lemma lemma-spaces-fibred-in-groupoids (canonical tag 0D1E). cited at proof line 4796. The corresponding fact for the spaces stack.
- Lemma spaces-morphisms-lemma-dimension-fibre-after-base-change (canonical tag 04NS). cited at proof line 4809. Relative dimension <= 1 after base change.

Source reference. Original equation 0D50: sources/stacks-chow/tags.txt, source lines 16470–16470. Cited at proof line 4793: the embedding.

Source reference. Original lemma 0D1E: sources/stacks-chow/tags.txt, source lines 16344–16344. Cited at proof line 4796: the corresponding fact.

Source reference. Original lemma 04NS: sources/stacks-chow/tags.txt, source lines 6090–6090. Cited at proof line 4809: relative dimension <= 1 preserved.

Lemma 0.2. *The category $\mathcal{C}urves$ is fibred in groupoids over Sch_{fppf}.*

Proof. Using the embedding (0D50), the description of the image, and the corresponding fact for $\mathcal{S}paces'_{fp,flat,proper}$ (Lemma 0D1E) this reduces to the following statement: Given a morphism

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S \end{array}$$

in $\mathcal{S}paces'_{fp,flat,proper}$ (recall that this implies in particular the diagram is cartesian) if $X \to S$ has relative dimension ≤ 1, then $X' \to S'$ has relative dimension ≤ 1. This follows from Morphisms of Spaces, Lemma 04NS. $\square$

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